

An Empirical Study of the DSOS/SDSOS Hierarchy

Tractable inner approximations for dense polynomial optimization

Diane Tchuindjo

May 2026

6.7230 — Algebraic Techniques and Semidefinite Optimization

The Trade-Off

SOS minimization of degree- $2d$ p : find $Q \succeq 0$ with $p(x) - \gamma = z_d(x)^\top Q z_d(x)$, then γ certifies a lower bound.

- This is an **SDP** of size $m = \binom{n+d}{d}$ making it expensive, and dense instances grow fast.
- **DSOS** / **SDSOS**: replace PSD by diagonally dominant (DD) / scaled diagonally dominant (SDD) cones $\Rightarrow$ an **LP** / **SOCP**.

The promise and the risk

Cheaper conic problems, but $DD \subsetneq SDD \subsetneq PSD$ are *strict* inner approximations, so they can give weak bounds, or fail outright, on instances that are easy for SOS.

When is that trade actually safe?

Proposal feedback raised four points. The final study is deliberately narrowed to answer them cleanly rather than survey broadly:

1. **Solver & algorithm dependence** → explicit CLARABEL/SCS check; SDSOS+ basis refinement.
2. **Reusable instances** → a saved, regenerable instance library as a first-class object.
3. **Random-ensemble dependence** → *two* contrasting ensembles, never averaged.
4. **Theory** → tie results to PSD-cone approximation lower bounds (Song–Parrilo).

One controlled benchmark of dense unconstrained polynomials + two diagnostics.

For a cone $\mathcal{C} \subseteq \mathbb{S}^m$:

$$\gamma_{\mathcal{C}}(p) = \sup_{\gamma, Q} \gamma \quad \text{s.t.} \quad p(x) - \gamma = z_d(x)^\top Q z_d(x), \quad Q \in \mathcal{C}.$$

$$\text{DD} \subsetneq \text{SDD} \subsetneq \text{PSD} \quad \implies \quad \gamma_{\text{DSOS}} \leq \gamma_{\text{SDSOS}} \leq \gamma_{\text{SOS}}.$$

- DD: polyhedral $\Rightarrow$ LP. SDD: factor-width-two ($O(m^2)$ embedded 2×2 blocks) $\Rightarrow$ SOCP.
- Reference = SOS. Performance = relative gap $\text{relgap}_{\mathcal{C}} = \frac{\gamma_{\text{SOS}} - \gamma_{\mathcal{C}}}{\max\{1, |\gamma_{\text{SOS}}|\}}.$
- Infeasible solves contribute only to *success rate*.

Design: Two Ensembles

DD-generated (easy_dsos)

Gram matrix DD in the monomial basis $\Rightarrow$ a DSOS certificate exists by construction.
DSOS-friendly.

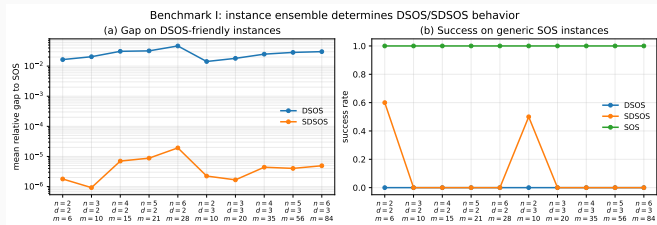
Generic-SOS (hard_sos)

$Q = A^\top A$ with Gaussian A , controlled rank and conditioning $\Rightarrow$ SOS feasible by construction; no DD/SDD structure forced.
Stress test.

Grid: $n \in \{2, \dots, 6\}$, $d \in \{2, 3\}$, 10 seeds/cell $\Rightarrow$ 100 instances each.

Reproducibility: every instance saved as JSONL *before* solving (basis, coeffs, seed). Same polynomial re-solved across cone / solver / basis.

Benchmark I: The Ensemble Decides Everything



DD-generated: DSOS, SDSOS, SOS
all 100/100; SDSOS gap to SOS
 $\sim 10^{-6}$.

Generic-SOS: SOS 100/100; **DSOS**
0/100; fixed-basis SDSOS only 11/100
(smallest two cells only).

Takeaway: “performance of SDSOS”
is undefined without naming the
ensemble. (Timing: at these small
dense sizes SOS is often *fastest*
through CVXPY — implementation
effect, not an asymptotic claim.)

Why Does Fixed-Basis SDSOS Fail? Two Hypotheses

H1: cone is too weak

No SDD certificate exists for these polynomials in *any* basis. SDSOS is fundamentally insufficient $\Rightarrow$ nothing to do.

H2: basis is bad

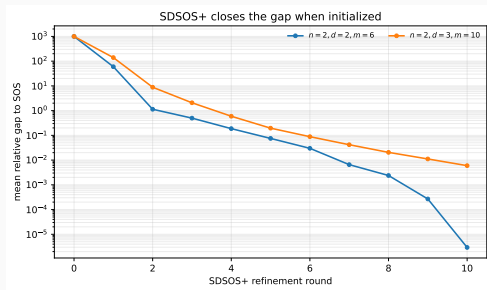
An SDD certificate *does* exist, just not in the monomial basis. Rewrite the problem in different coordinates and it appears.

Key

PSD-ness is invariant under congruence $Q \mapsto U^T Q U$ so basis is irrelevant for SOS. But **diagonal dominance is coordinate-dependent**, so SDD-feasibility *moves* with the basis. H2 is therefore possible in principle.

SDSOS+ idea: iterate SDSOS, using the *factor of the current solution* as the next basis. Each round re-centers the SDD cone on the certificate found so far, opening room to push γ further. Stays an SOCP throughout[Ahmadi & Hall]. **Catch:** needs round 0 to be feasible to start.

Benchmark II: Most of the Loss is the Basis



SDSOS+ basis refinement: when it can be *initialized*, the gap drops several orders of magnitude.

Oracle-basis diagnostic ($m \leq 20$, generic-SOS): solve SOS, factor its Gram, re-solve SDSOS in that basis.

<i>n</i>	<i>d</i>	<i>m</i>	Fixed	SOS-basis
			SDSOS	SDSOS
2	2	6	0.60	1.00
3	2	10	0.00	0.90
4	2	15	0.00	0.60
2	3	10	0.50	0.90
3	3	20	0.00	0.50
total			11/50	39/50

Failure is **partly initialization**, not pure cone weakness, but SDSOS+ cannot start unless round 0 is already feasible.

Solver Sensitivity: Is the story a CLARABEL Artifact?

Re-run a slice ($n \in \{2, 3, 4\}$, $d = 2$, 5 seeds) under **SCS**. This is a structurally different solver (first-order splitting, not interior-point).

Ensemble	Cone	CLARABEL		SCS	
		Succ./Opt.	Gap	Succ./Opt.	Gap
DD-gen.	SDSOS	1.00 / 1.00	2.1×10^{-6}	1.00 / 0.93	-4.7×10^{-6}
Generic	DSOS	0.00 / 0.00	—	0.00 / 0.00	—
Generic	SDSOS	0.20 / 0.13	1.5×10^3	1.00 / 0.00	7.7×10^4
Generic	SOS	1.00 / 1.00	5.0×10^{-6}	1.00 / 1.00	9.4×10^{-8}

Succ.: status optimal or optimal_inaccurate. **Opt.:** optimal only — the trustworthy subset.

Solver Sensitivity: What the Table Tells Us

1. **The story is solver-robust.** Easy/hard split holds under both solvers; DSOS is 0/15 on generic regardless of engine.
2. **Nominal success $\neq$ feasibility.** On hard SDSOS, SCS reports **1.00 success but 0.00 reliable** — all 15 runs labeled `optimal_inaccurate`. The solver is flagging its own imprecision.
3. **The negative gap is a tell, not a paradox.** $\text{SDD} \subseteq \text{PSD}$ forces $\gamma_{\text{SDSOS}} \leq \gamma_{\text{SOS}}$ *in exact arithmetic*, so a negative gap is mathematically impossible. SCS's -4.7×10^{-6} is the iterate crossing the cone boundary within the solver's own feasibility tolerance. This tells us that “optimal” can mean “feasible to tolerance,” not feasible.

Why this is not an accident: the theory

DSOS/SDSOS optimize over the factor-width-1/2 cones, i.e, the *duals* of the $k=1, 2$ PSD closures (Boman et al.; Ahmadi–Majumdar: $k=1 \rightarrow \text{LP}$, $k=2 \rightarrow \text{SOCP}$).

Song & Parrilo (Math. Program. 2022)

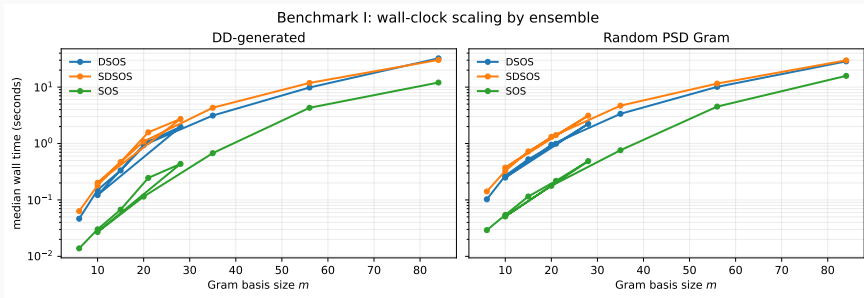
Any **data-oblivious** $\mathbf{S}_+^k$ -approximation S of $B_H(\mathbf{S}_+^n)$ has extension complexity $\text{xc}_{\mathbf{S}_+^k}(S) \geq \exp(C \min\{\sqrt{n}, n/k\})$ — so no polynomial number of fixed $k \times k$ blocks gives a constant-factor approximation once $k = o(n/\log^2 n)$.

- SDSOS = factor width 2, DSOS = factor width 1, so the bound covers them **by duality**
- The generic-SOS collapse is *consistent with* this (cf. the $m^{-1/2}$ vs. m^{-1} heuristic), its not a derivation of it.
- No tension with Benchmark II: the bound is **worst-case and data-oblivious**; the paper itself leaves the data-adaptive regime open.

Conclusion

1. **Ensemble dominates.** Near-lossless on DD-generated; near-useless on generic-SOS. An unqualified claim about SDSOS is not well posed.
2. **Most loss is basis, not cone.** Oracle basis: $11/50 \rightarrow 39/50$; SDSOS+ recovers orders of magnitude when initialized. Open problem moves to *basis discovery without an SDP oracle*.
3. **Status $\neq$ feasibility.** Conclusions are solver-robust, but inaccurate statuses must not be trusted.
4. **Consistent with theory.** The failures are the expected shadow of a proven PSD-approximation lower bound.

Implementation Timing



Median wall time vs. m , including CVXPY canonicalization. At the tested dense sizes SOS is often fastest within this common high-level stack — an implementation-specific sanity check, *not* an asymptotic LP/SOCP-vs-SDP comparison.

Backup: Definitions

$Q \in \mathbb{S}^m$ symmetric. Three nested cones:

Diagonally dominant (DD)

$Q_{ii} \geq \sum_{j \neq i} |Q_{ij}|$ for every row i .

Polyhedral cone $\Rightarrow$ optimizing over DD is an **LP**.

Generators: $e_i e_i^\top$ and $(e_i \pm e_j)(e_i \pm e_j)^\top$.

Scaled diagonally dominant (SDD)

There exists a positive diagonal D with $DQD \in \text{DD}$.

Equivalently: Q is a sum of 2×2 PSD blocks (*factor width 2*) $\Rightarrow$ **SOCP**.

Positive semidefinite (PSD)

$v^\top Q v \geq 0$ for all $v \in \mathbb{R}^m$; all eigenvalues ≥ 0 .

Optimizing over PSD is an **SDP**.

Backup: Hard-Ensemble Construction

Per cell of the (n, d) grid; $m = \binom{n+d}{d}$:

1. Sample $A \in \mathbb{R}^{r \times m}$ with i.i.d. $\mathcal{N}(0, 1)$ entries, $r = \lceil \text{rank_frac} \cdot m \rceil$.
2. Row-rescale: $A_i \leftarrow A_i / s_i$ with s_i geometrically spaced in $[1, \text{condition}]$.
3. Form $Q = A^\top A$, symmetrize $Q \leftarrow \frac{1}{2}(Q + Q^\top)$.
4. $p(x) := z_d(x)^\top Q z_d(x)$; add $\text{linear_scale} \cdot \mathcal{N}(0, 1)$ noise to linear coefficients; add constant shift.

Backup: Extended Detail on Benchmark II

Both ask “is the failure a bad *basis* rather than a weak cone?” but through structurally different procedures.

SDSOS+ (iterative, no SOS access)

Round 0: solve SDSOS in monomial basis $\rightarrow$ Gram $Q^{(0)}$.

Round $k + 1$: factor $Q^{(k)} = R_k^\top R_k$; re-solve SDSOS in basis $y_{k+1} = R_k y_k$.

Iterate up to 10 rounds. Each step uses only *SDSOS's own previous output*. Requires round 0 to be feasible.

Oracle basis (one-shot, uses SOS)

Solve SOS $\rightarrow$ Gram Q_{SOS}^* ; factor $Q_{\text{SOS}}^* = R^\top R$; solve SDSOS *once* in basis $R z_d$. *diagnostic*.

Backup: Diagnostic Conclusion

Same 50 hard instances ($m \leq 20$). Two complementary readings:

Instances	SDSOS+ says	Oracle says
11 where round-0 SDSOS succeeds	gap collapses by $\sim 10^3$ – 10^6 over 10 rounds	also succeeds, gap $\sim 10^{-7}$
39 where round-0 SDSOS fails	can't start; silent	succeeds on 28; fails on 11

- **SDSOS+ alone:** shows basis refinement *works* when it can be initialized.
- **Oracle alone:** shows a good basis *exists* for most instances, even those SDSOS+ can't reach.
- **Together:** basis, not cone, is the bottleneck. Residual 11/50 oracle failures: genuine structural gap, rank-deficient SOS factor, and numerical conditioning at $m = 20$.